Chapter 6: Consumption, Fiscal Policy and Ricardian Equivalence

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1 Overview

This chapter is the first building block in constructing a modern macro model. In some respect, the structure of future chapters can be thought of as an onion, each add an extra layer of complexity. At the core of it all is the Consumption Euler Equation we present in this chapter. Some economists, probably without exaggeration, have called it the most important equation in macroeconomics.

The big picture is to use micro-foundations to derive a consumption function. Consumption behavior is at the heart of all macro models. This is because consumption corresponds to about 65 percent of GDP and thus the most fundamental part of aggregate spending.

To get some perspective on the modern theory of consumption, it is useful to get perspective on what it replaced. The old Keynesian consumption function from the IS-LM model is:

$$C_t = \alpha_0 + \alpha(Y_t - T_t),$$

where $0 < \alpha < 1$ is the marginal propensity to consume and α_0 is a constant. This consumption function is counterfactual because it says that consumers always consume a fixed fraction of their annual income. However, if a woman won a lottery ticket of 1 million, this would cause a very different effect than if her salary per year increased by 1 million. In the first case, she will save a lot of the one million (perhaps only spending corresponding to the interest she earns), while in the latter case, she will spend a much larger fraction, since her lifetime income has permanently increased by one million per year.

This led people to conjecture that consumption should not be a function of "current" income but instead of "permanent income", an idea that Milton Friedman was awarded Nobel Prize for. The bottom line is that we will derive an expression such as:

$$C_t = \alpha Y_t^{per} + \text{other terms}, \tag{1}$$

where Y_t^{per} will be a measure of permanent income of a form that we will derive.

An interesting aspect of this derivation is that Y_t^{per} will not depend upon taxes, a proposition often called Ricardian Equivalence. The Ricardian Equivalence says that the time path of taxes does not matter for consumption or interest rate. The reason is that in the model we will derive, if taxes are cut today, then people will save the tax cut to pay for taxes in the future. As a result, there is no effect of consumption on the interest rate. Both will, however, depend upon government spending G_t.

While the Ricardian Equivalence result is extreme and will fail to hold under variety of extensions, the big picture here is to build up behavioral relationships such as (1) from micro-foundations, which will reveal that expectations about the future play a fundamental role. While the model presented here is very stylized, the models we will see in the rest of the class will build directly on top of it, just relaxing one extreme assumption at a time towards more realism, adding another layer of complexity.

2 Basic Setup

The Households

There is a representative household solves the following problem:

$$\max_{C_t, b_t} E_t \sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\sigma^{-1}}}{1-\sigma^{-1}}$$

s.t.

$$b_t = (1 + r_{t-1})b_{t-1} + Y_t - C_t - T_t \tag{2}$$

where b_t is one-period risk-free bond, Y_t is income, C_t consumption, T_t taxes and E_t is an expectation operator.

The Government

The government collects taxes T_t and decides on spending G_t. All income is either consumed by the agent or the government, implying the following *resource constraint*:

$$Y_t = C_t + G_t \tag{3}$$

Remark 1: The household budget constraint (2) and the resource constraint (3) together imply a budget constraint for the government (and vice versa). To see this, just plug (3) into (2) to obtain the government budget constraint:

$$b_t = (1 + r_{t-1})b_{t-1} + G_t - T_t \tag{4}$$

This is relevant because, when we define equilibrium, we do so by listing the set of equations that need to be satisfied. If you state the budget constraint of the household (2) and the resource constraint (3), then stating the government budget constraint (4) is redundant. You can similarly state the government budget constraint and the resource constraint; then the household budget constraint is redundant.

Remark 2: Since all households are the same, then in equilibrium all the assets held by the households in this economy will be government debt. We assume that there is a large number of households, so that they take the interest rate as given. We normalize the number of households to 1.

Remark 3: A more precise notation would state that the household is choosing b_t^d where d stands for bonds "demanded." The government then chooses the supply b_t^s. In equilibrium, then: $b_t^d = b_t^s = b_t$.

Remark 4: We only consider a one-period bond being traded for now. Also, while we keep track of uncertainty via the expectation operator, we do not formally discuss issues related to uncertainty yet. Both issues are addressed in the next chapter.

The Household's Program

What does the household do? To characterize it, we write a Lagrangian:

$$\mathcal{L}_0 = E_0 \sum_{t=0}^{\infty} \beta^t \left\{ \frac{C_t^{1-\sigma^{-1}}}{1-\sigma^{-1}} + \phi_t \left[b_t - (1+r_{t-1}) b_{t-1} - Y_t + C_t + T_t \right] \right\}$$

First-order conditions (FOCs):

$$\frac{\partial \mathcal{L}_0}{\partial C_t} = \beta^t C_t^{-\sigma^{-1}} + \beta^t \phi_t = 0 \tag{5}$$

$$\frac{\partial \mathcal{L}_0}{\partial b_t} = \beta^t \phi_t - \beta^{t+1}(1+r_t) E_t \phi_{t+1} = 0 \tag{6}$$

Combing the FOCs yields the consumption Euler Equation:

$$C_t^{-\sigma^{-1}} = \beta(1+r_t) E_t C_{t+1}^{-\sigma^{-1}} \tag{7}$$

The household also needs to satisfy a transversality condition (which we derive in the next chapter):

$$\lim_{T\to\infty} E_t \beta^T C_T^{-\sigma^{-1}} b_T = 0 \tag{8}$$

The transversality condition stated in (8) applies if we interpret the model as applying to the case in which markets are complete, so that any arbitrary stream of income can be priced and traded (even if we assume here that only one-period debt is traded in equilibrium). The results are exactly the same if we assume instead that markets are incomplete in the sense that the only assets that can be traded is one-period debt. The only difference in this case, however, is that the transversality condition becomes:

$$\lim_{T\to\infty} \beta^T C_T^{-\sigma^{-1}} b_T = 0 \tag{9}$$

which differs from 8 only in that there is no expectation operator. The interpretation of this condition is that it prevents explosive paths for debt, not only in expectations, but for any realization.

The Government's Program

We do not assume, for now, that fiscal policy is set to optimize an objective. Instead, it is just a function:

$$T_t = T(Y_t) \tag{10}$$

$$G_t = G(Y_t) \tag{11}$$

defining the sequences $\{T_t\}$ and $\{G_t\}$. Because the only source of uncertainty is output variability, there is no loss in generality from not including either C_t or r_t as elements of the functions, for these are functions of the endowment as well.

What is a solution in this model? It describes how each of the endogenous variables depends on the exogenous driving forces (here the only exogenous variable is Y_t). It is a useful practice when analyzing models to explicitly define an equilibrium (see below). This amounts to little more than a laundry list of equations that need to be satisfied in a particular model for a solution to be meaningful.

Definition 1. *An equilibrium consistent* with risk-free government debt *is a collection of processes* $\{b_t, r_t, C_t\}$ *for* $t \geq 0$ *that satisfy (3),(4), (7) and (8),*

given a process for $\{Y_t\}$*, a policy specification for* $\{T_t, G_t\}$ *given by (10) and (11) and an initial condition* b_0.

It can be shown that the list of conditions listed above are necessary and sufficient for an equilibrium consistent with household maximization. We could have replaced the government budget constraint with the household budget constraint, as one implies the other, given the resource constraint (3).

We define sustainable fiscal policy as a policy specification for the function $T(.)$ and $G(.)$ so that public debt never explodes, i.e. it guarantees that the transverality condition of the household is always satisfied.

Definition 2. *A sustainable fiscal policy is a specification for* $T(Y_t)$ *and* $G(Y_t)$*, given* b_0*, so that (8) is satisfied for any* $\{Y_t\}$.

Example 1: Consider the example in which $b_{-1} = 0$ and the budget is balanced at all times, that is, $T_t = G_t$ at all times. This policy is then clearly an example of a sustainable fiscal policy (to see this, just note that the transversality condition is satisfied for any path of r_t, given that $b_t = 0 \; \forall t$).

Remark 4: Assuming a sustainable fiscal policy puts restriction on the functions $T(.)$ and $G(.)$. Also, note that we need $T(.)$ and $G(.)$ to be sustainable for the debt to be risk-free.

3 Ricardian Equivalence

Here, we state the Ricardian Equivalence in a relatively abstract way. The reason why we temporarily do it in this way will hopefully be a little bit more clear later.

Proposition 3 (Ricardian Equivalence)**.** *Assume that fiscal policy is sustainable. In this case, the determination of* C_t *and* r_t *is independent of the specification of the function* $T(Y_t)$.

Proof. We can simplify the list of conditions listed in our definition of an equilibrium to:

$$(Y_t - G(Y_t))^{-\sigma^{-1}} = \beta(1 + r_t)E_t(Y_{t+1} - G(Y_{t+1}))^{-\sigma^{-1}} \tag{12}$$

$$b_t = (1 + r_{t-1})b_{t-1} + G(Y_t) - T(Y_t) \tag{13}$$

$$\lim_{T\to\infty} E_t\beta^T(Y_T - G(Y_T))^{-\sigma^{-1}}b_T = 0 \tag{14}$$

For a given function $G(Y_t)$ and a stochastic process for Y_t, the first equation pins down r_t independently of the function $T(Y_t)$, i.e. we can define any function for

$T(.)$, as long as the policy is sustainable, which ensures that the transversality condition is satisfied. Ricardian equivalence implies that the time path of taxes does not matter for C_t or r_t, as long as it is sustainable, which is needed for the existence of risk-free debt equilibrium. □

What does Ricardian equivalence say? In this simple setup it says that the short term real interest rates does not depend on taxes. More generally, it says that the consumption of the household does not depend on taxes today. This can be shown a bit more clearly by using the budget constraint of the household and the expectation operator as we do below.

Ricardian equivalence does not say that the interest rate and aggregate consumption determination is independent of G_t. Clearly, as G_t increases, by the resource constraint C_t will need to decline and the interest rate will adjust so as to satisfy the consumption Euler Equation. You would be surprised how many people get confused on this point and appear to think Ricardian Equivalence implies that the determination of G_t is irrelevant. Clearly, it is not.

4 Showing Ricardian Equivalence by Forwarding the Budget Constraint

What we aim to show here is that the maximization problem of the household, i.e. how it determines consumption, can be stated without any references to taxes. The implication, of course, is that taxes are irrelevant for consumption decisions. This alternative way of showing the Ricardian Equivalence gives a bit clearer window into the logic of the Ricardian proposition in the last section.

Remark 5: First, a little bit of notation. The *discount factor* tells you the price of one (real) dollar tomorrow in terms of today's price, as further discussed in the next chapter:

$$R_{t,t+1} \equiv \frac{1}{1+r_t}$$

Similarly, the price of one dollar at time T in today's price is:

$$R_{t,T} \equiv \frac{1}{1+r_t}\frac{1}{1+r_{t+1}}...\frac{1}{1+r_{T-1}}$$

or:

$$R_{t,T} \equiv \frac{1}{\prod_{s=t}^{T-1}(1+r_s)}$$

We use the notation that $R_{t,t} \equiv 1$, that is, the price of one dollar today in today's prices is – well – one dollar. Observe that:

$$R_{t,T} = R_{t,J} * R_{J,T}$$

where $t < J < T$. Alternatively, one can write:

$$R_{t,T} = \prod_{s=t+1}^{T} R_{s-1,s}$$

Remark 6: It is a little bit more convenient to define the budget constraint in terms of a variable V_t, defined as the beginning wealth in period t carried from the last period:

$$V_t \equiv (1 + r_{t-1})b_{t-1}$$

The transversality condition can alternatively be stated as (see Appendix for derivation):

$$\lim_{T\to\infty} E_t R_{t,T} V_T = 0 \tag{15}$$

It is straightforward to write the household budget constraint in terms of this variable as:

$$V_{t+1} = (1 + r_t)(V_t + Y_t - C_t - T_t)$$

or, we can alternatively state this as:

$$V_t = \frac{V_{t+1}}{1 + r_t} - Y_t + C_t + T_t$$

You can forward the period t budget constraint to yield:

$$\begin{aligned} V_t &= \frac{V_{t+1}}{1 + r_t} - Y_t + C_t + T_t \\ &= \frac{1}{1 + r_t}[\frac{V_{t+2}}{1 + r_{t+1}} - Y_{t+1} + C_{t+1} + T_{t+1}] - Y_t + C_t + T_t \\ &= \frac{1}{1 + r_t}\frac{V_{t+2}}{1 + r_{t+1}} - \frac{1}{1 + r_t}Y_{t+1} + \frac{1}{1 + r_t}C_{t+1} + \frac{1}{1 + r_t}T_{t+1} - Y_t + C_t + T_t \end{aligned}$$

or:

$$V_t = R_{t,T}V_T + \sum_{s=t}^{T-1} R_{t,s}(T_s + C_s - Y_s)$$

Using the transversality condition, we can write:

$$\sum_{s=t}^{\infty} R_{t,s} C_s = V_t + \sum_{s=t}^{\infty} R_{t,s}(Y_s - T_s), \tag{16}$$

which is the intertemporal budget constraint of the household.
We can similarly rewrite the budget constraint of the government as:

$$V_t = \frac{V_{t+1}}{1 + r_t} - G_t + T_t$$

and forward it to yield:

$$\sum_{s=t}^{\infty} R_{t,s} G_s + V_t = \sum_{s=t}^{\infty} R_{t,s} T_s \tag{17}$$

We can substitute (17) into (16) to obtain:

$$\sum_{s=t}^{\infty} R_{t,s} C_s = \sum_{s=t}^{\infty} R_{t,s} Y_s - \sum_{s=t}^{\infty} R_{t,s} G_s \tag{18}$$

What does this say? We have rewritten the intertemporal budget constraint of the household, in order to show that the consumption choices of the household will not be affected by the timing of taxes because they do not appear in the budget constraint, which limits the choices taken by the household. This is the way in which the Ricardian Equivalence is often stated. What happens when the government cuts taxes? The household increases its savings one to one to meet its future tax liability.

5 Solving for the Consumption Function

Remember the big picture? It was to derive a consumption function, i.e. show how consumption can be expressed as a function of permanent income. This is what we now do.

An equilibrium is defined by the following equations:

$$C_t^{-\sigma^{-1}} = \beta(1 + r_t) E_t C_{t+1}^{-\sigma^{-1}} \tag{19}$$

$$\frac{V_{t+1}}{1 + r_t} = V_t + Y_t - C_t - T_t \tag{20}$$

$$Y_t = C_t + G_t \tag{21}$$

and:

$$\lim_{T\to\infty} E_t R_{t,T} V_T = 0, \tag{22}$$

given a path for the policy functions $\{T_t\}$ and $\{G_t\}$ and for the exogenous process $\{Y_t\}$.

Consider a steady state of this model where the forcing variables are given by $Y_t = \bar{Y}$ and $G_t = \bar{G}$ and consider a fiscal policy such that $V_t = \bar{V}$. The steady state interest rate and taxes are then given by:

$$1 + \bar{r} = \frac{1}{\beta} \tag{23}$$

$$\bar{T} = \frac{\bar{r}}{1+\bar{r}}\bar{V} + \bar{G} \tag{24}$$

A linear approximation of the Euler Equation, as shown in Chapter 2, is now given by (here $\hat{C}_t \equiv \frac{C_t - \bar{C}}{\bar{C}}$):

$$\hat{C}_t = E_t \hat{C}_{t+1} - \sigma \hat{r}_t \tag{25}$$

The budget constraint is:

$$\bar{V}\beta\hat{V}_{t+1} = \bar{V}\hat{V}_t + \bar{Y}\hat{Y}_t - \bar{C}\hat{C}_t - \bar{T}\hat{T}_t + \beta\bar{V}\hat{r}_t,$$

which can be forwarded as before to obtain, using the government constraint, the resource constraint and taking conditional expectation on both sides:

$$\bar{C}\sum_{T=t}^{\infty}\beta^{T-t}E_t\hat{C}_T = \bar{Y}\sum_{T=t}^{\infty}\beta^{T-t}E_t\hat{Y}_T - \bar{G}\sum_{T=t}^{\infty}\beta^{T-t}E_t\hat{G}_T \tag{26}$$

This equation, together with the consumption Euler equation (25) and some additional derivation (which is explained a little bit in the Appendix), can be solved to yield:

$$\bar{C}\hat{C}_t = (1-\beta)E_t\left[\sum_{T=t}^{\infty}\beta^{T-t}(\bar{Y}\hat{Y}_T - \bar{G}\hat{G}_T)\right] - \sigma\bar{C}\beta\sum_{T=t}^{\infty}\beta^{T-t}E_t\hat{r}_T \tag{27}$$

I have chosen here to use the "hat" notation as we will be using it in the rest of the class. This is one example, however, when the expression is prettier without this notation, i.e. we can write actual consumption as:

$$C_t = (1-\beta)E_t\left[\sum_{T=t}^{\infty}\beta^{T-t}(Y_T - G_T)\right] - \sigma\beta\sum_{T=t}^{\infty}\beta^{T-t}E_t r_T \tag{28}$$

Here we see that, up to a first-order approximation, consumption is a fixed fraction $(1-\beta)$ of the present value of national income net of government purchases, but that it also depends negatively on the interest rate. This has a

similar flavor to the old Keynesian consumption function, but is different in several respects. For example, taxes are irrelevant and it is the net present value of income that matters, not the flow value for income. The net present value of income corresponds to the permanent income of the household. We also see the interest rate explicitly affecting spending, while in the most simple form of the Keynesian consumption function it does not.

6 Appendix

6.1 Derivation to get (15)

This proof is taken from Woodford (2003), and is relegated to the Appendix, as the starting point depends on (29) which we derive in next chapter.

From (8) we want to obtain (15). As we see formally in next chapter then

$$\frac{C_t^{-\sigma^{-1}}}{C_{t+1}^{-\sigma^{-1}}} = \beta(1+r_t) = \frac{\beta}{R_{t,t+1}}, \tag{29}$$

Rearranging:

$$C_{t+1}^{-\sigma^{-1}} = \frac{R_{t,t+1}}{\beta} C_t^{-\sigma^{-1}}$$

Substitute it in (8):

$$\lim_{T\to\infty} E_t \beta^T C_T^{-\sigma^{-1}} b_T = 0$$

$$\lim_{T\to\infty} E_t \beta^T \frac{R_{T-1,T}}{\beta} C_{T-1}^{-\sigma^{-1}} b_T = 0$$

$$\lim_{T\to\infty} E_t \beta^{T-1} R_{T-1,T} \frac{R_{T-2,T-1}}{\beta} C_{T-2}^{-\sigma^{-1}} b_T = 0$$

...

$$\lim_{T\to\infty} E_t R_{T-1,T} ... R_{t,t+1} C_t^{-\sigma^{-1}} b_T = 0$$

We can take $C_t^{-\sigma^{-1}}$ out of the expectation and of the limit. At the same time, within the expectation, we can multiply and divide by $(1+r_t)$ and obtain:

$$\lim_{T\to\infty} E_t R_{t,t+1} ... R_{T-1,T} \frac{1+r_t}{1+r_t} b_T = 0$$

$$\lim_{T\to\infty} E_t R_{T,T+1} V_{T+1} = 0$$

Finally, the limit will apply also for T, hence:

$$\lim_{T\to\infty} E_t R_{T-1,T} V_T = 0$$

6.2 Derivation to get (27)

We start with the intertemporal budget constraint:

$$\bar{C}\sum_{T=t}^{\infty}\beta^{T-t}E_t\hat{C}_T = \bar{Y}\sum_{T=t}^{\infty}\beta^{T-t}E_t\hat{Y}_T - \bar{G}\sum_{T=t}^{\infty}\beta^{T-t}E_t\hat{G}_T$$

To get to (27), we start by rewriting the Euler Equation, (25):

$$\hat{C}_t = E_t\hat{C}_{t+1} - \sigma\hat{r}_t$$

Now forward this equation and apply the conditional expectation:

$$E_t\hat{C}_{t+1} = \hat{C}_t + \sigma\hat{r}_t$$

Do the same for $E_t\hat{C}_{t+2}$:

$$E_t\hat{C}_{t+2} = E_tC_{t+1} + \sigma E_t\hat{r}_{t+1} = \hat{C}_t + \sigma\hat{r}_t + \sigma E_t\hat{r}_{t+1}$$

Do the same for $E_t\hat{C}_{t+3}$ – you can see the pattern summarized in (30):

$$E_t\hat{C}_{t+3} = E_tC_{t+2} + \sigma E_t\hat{r}_{t+2} = \hat{C}_t + \sigma\hat{r}_t + \sigma E_t\hat{r}_{t+1} + \sigma E_t\hat{r}_{t+2}$$

More generally:

$$E_t\hat{C}_T = \hat{C}_t + \sigma\sum_{s=t}^{T-1}E_t\hat{r}_s \tag{30}$$

We can now substitute this into $\bar{C}\sum_{T=t}^{\infty}\beta^{T-t}E_t\hat{C}_T$, which is a little tricky, and get the solution (the easiest way is to write out the sums explicitly).

7 Connection to the literature

My first job out of grad school was at the research department of the IMF. At the IMF we often had to assess whether fiscal policy in various countries was "sustainable." This boiled down to computing whether the budgetary plans of the government led to explosive debt or not. This, essentially, is how I define sustainable policy here. The monetary literature contains a similar definition, which we will get back to once we introduce monetary policy explicitly. Woodford, for example, calls sustainable fiscal policy "Ricardian." If the functions $T(.)$ and $G(.)$ are not consistent with sustainability Woodford terms them "non-Ricardian." Leeper (1992), in contrast, calls fiscal policy that is sustainable "active," while policy that does not guarantee sustainability is called "passive." We will discuss Leeper's paper and the connection between unsustainability and the so-called fiscal theory of the price level in Chapter 9.

8 Problems

Consider a representative household as in class

$$\max_{C_t, b_t} E_t \sum_{t=0}^{\infty} \beta^t \frac{(C_t - H_t)^{1-\sigma}}{1-\sigma}$$

s.t.

$$b_t = (1 + r_{t-1})b_{t-1} + Y_t - C_t - T_t$$

where C_t is consumption, r_t the real interest rate, b_t is debt, Y_t is output, C_t is consumption and T_t is taxes. The coefficient β satisfies $0 < \beta < 1$ and $\sigma > 0$. The variable H_t denotes the stock of "habits" of the household, which the household *takes as given* (or exogenous), a new feature relative to the model in class. This form of "consumption habit" is common in the macroeconomic literature: The idea is that people do not care about the absolute level of consumption but how much they consume relative to "what they are used to" or some "habit" that may be formed over time.

i. Write down the first order conditions of the household. Note that H_t is not a decision variable here and the household will treat it as some exogenous shock.

ii. Let us assume that $H_t = \gamma C_{t-1}$. Assume that $\{G_t\}$ is exogenous. Assume that fiscal policy is sustainable in the sense defined in class, i.e. that taxes are set so that debt is always finite according to some function $T(Y_t, G_t)$. Define the equilibrium. (When I say "define the equilibrium" I am just asking you to write down the conditions that need to be satisfied in this economy).

iii. Find the steady state of the model assuming $Y_t = \bar{Y}$. Do a linear approximation and derive an approximate equilibrium.

iv. Assume that Y_t follows the AR(1) processes:

$$\hat{Y}_t = \rho \hat{Y}_{t-1} + \epsilon_t$$

and that $\hat{G}_t$ follows the process:

$$\hat{G}_t = \alpha \hat{Y}_t + \rho \hat{G}_{t-1} + u_t$$

Compute the impulse response for the real interest rate in response to both ϵ_t and u_t.

v. What is the impulse response function for taxes? ***Hint:*** This model exhibits Ricardian Equivalence!